

Ibrahim Ali

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EDUCATION

- Vellore Institute of Technology** Chennai, India
B.Tech in Computer Science and Engineering; CGPA: 8.7 Sep 2020 – May 2024

EXPERIENCE

- Air India Limited** Gurugram, India
Software Engineer August 2025 – Present
 - Led the modernization of a high-throughput event-driven delivery platform processing **3,500+ flight events/day** and **500,000+ passenger notifications/day**, replacing a polling-based scheduler with a real-time architecture that significantly improved operational efficiency and system responsiveness.
 - Engineered Air India's delivery platform into a **distributed microservices architecture** using **Azure Service Bus, and asynchronous messaging**, scaling throughput to **1M+ daily communications** and sustaining **50,000+ requests/hour** under peak demand.
 - Optimized the communication platform by replacing vendor-managed services with **AWS SES**-based in-house delivery and shifting workloads to **serverless** and **containerized** runtimes on **Azure Functions** and **AKS**, achieving **60% infrastructure cost reduction** through scalable cloud-native design.
 - Contributed to a **scheduling engine** for **100,000+ booking transactions/day** and **400,000+ time-based workflows**, combining streaming ingestion with periodic schedule reconciliation to trigger configurable delivery pipelines across customer journeys.
 - Developed an automated **flight disruption processing framework** that ingests reservation updates, applies rule-based validation for delay and prepone scenarios, eliminating manual coordination and reducing operational expenses by **40%**.
 - Designed a **two-tier caching architecture** combining application-level **LRU in-memory caching** with **Redis-based distributed caching**, reducing access latency for high-read, low-change datasets and improving response times from **2 seconds to sub-200ms**.
 - Collaborated with cross-functional engineering teams to build an **MCP-based client-server platform** integrated with backend **Spring Boot** services, reducing production issue identification time from **30 minutes to under 5 minutes** through real-time system diagnostics and tool-driven access.
- Air India Limited** Kochi, India
Software Development Engineer Trainee Aug 2024 – July 2025
 - Engineered a scalable survey distribution pipeline processing **150,000+ burst records** via **Azure Event Hub**, leveraging **Azure Data Factory** for throughput orchestration and **Service Bus**-based callback handling to ensure reliable third-party API integration with retryable state management.
 - Implemented a **channel-prioritization strategy** centered on **WhatsApp and Email delivery**, reducing monthly communication costs by **60%+** while improving reachability and engagement across customer touchpoints.
 - Strengthened production engineering standards through **PR governance**, **SonarQube analysis**, automated unit testing, and **GitHub security scanning**, while building **Grafana observability dashboards** and streaming finalized records to **Databricks** for analytics-driven optimization.

PROJECTS

- Flight Disruption Management System (Prototype)** — Python, React.js, Claude Code, LangChain, PostgreSQL
 - Designed an **AI-driven 4-agent architecture** for disruption impact analysis, severity scoring, resolution generation, and decision orchestration.
 - Independently production-hardened workflows through **deterministic state transitions** across **5 conversation stages**, improving reliability and auditability.
 - Expanded coordination logic and memory systems to support **production outage resolution** through extra operational effort and sustained execution under disruption spikes.

ACHIEVEMENTS

- Dream Team Award (2025)** — For engineering excellence in mission-critical systems.
- High Flyer Award (2025, 2026)** — For sustained contributions to scalability and reliability.
- AI IMPACCT Champion Award (2024, 2025)** — For production outage resolution.

SKILLS

- Languages: Java, C++, Python, SQL, JavaScript
- Backend: Spring Boot, JUnit, REST APIs, SOAP APIs, Microservices, Event-Driven Systems
- Cloud & DevOps: Azure(Service Bus, Event Hub, App Service), AWS (S3, EC2, Lambda, CloudWatch), Kafka, Docker, Kubernetes, Grafana, Azure Pipelines
- Databases: PostgreSQL, MySQL, CosmosDB, Redis
- Foundations: Object-Oriented Design, Design Patterns, Data Structures, Distributed Systems